

Site of Photosynthesis

Photosynthesis takes place in chloroplasts. These are the green plastids, which occur in green parts of plants.

- Maximum number of chloroplasts are present in leaves.
- It is a double membranous organelle enclosing a liquid matrix called **stroma**.
- In the stroma, tiny, membranous sac-like lamellae present. These are called as **thylakoids**.
- Thylakoids are stacked over one another to form **grana**.
- Stroma is the site of dark reaction and thylakoids for light reaction.

Photosynthetic Pigments

Pigments are the organic molecules that absorb the light of specific wavelength in the visible region. Different types of pigments are:

Chlorophylls (Gk. *Chlor*–green, *phyll*–leaf)

These are the green photosynthetic pigments present in all photosynthetic autotrophic organisms. There are about ten types of chlorophylls-*a*, *b*, *c*, *d* and *e*, bacteria chlorophyll-*a*, *b*, *c* and *d* and bacteriovirdin. Chl-*a* is found in all the oxygen evolving organisms. It is the **primary photosynthetic pigment**.

Structure of Chlorophyll

The structure of chlorophyll consists of porphyrin head and phytol tail.

- The porphyrin head is made up of four pyrole rings linked together by methane groups, forming a ring system.
- In the centre of tetrapyrrole ring a divalent ion Mg^{2+} is present, which is complexed with the nitrogen atoms of four pyrole rings.
- The phytol tail is made up of 20 carbon alcohol and bound by ester linkage to the 4th pyrole ring.

Carotenoids

These are accessory pigments that help in photosynthesis in plants. Two major classes of carotenoids are carotenes (contain no oxygen) and xanthophylls (contain oxygen atom in their terminal ring).

Phycobilins

These pigments are found in red algae and blue-green algae, but not in higher plants. The phycoerythrin and phycocyanin are the two main phycobilins found in red algae and blue-green algae, respectively.

Mechanism of Photosynthesis

Photosynthesis is a two step process:

- The **first step** is dependent on light and responsible for accumulation of assimilatory power. It is called as **light reaction**.
- The **second step** is called **dark reaction** as light is not required for the purpose. It is responsible for CO_2 -fixation into carbohydrate.



- If vaseline is applied on both the surfaces of the leaf of a plant photosynthesis, transpiration and respiration in plant will get affected as all these processes require the direct contact between the leaf and atmosphere.
- Mushroom (white fungi) plant is not capable of manufacturing its own food.

RESPIRATION

It is the process of exchange of respiratory gases (O_2 and CO_2) between an organism and its environment. Respiration is the most important catabolic process, which involves the liberation of energy by oxidation and breakdown of food inside living cell.

On the basis of O_2 , respiration is of two types:

- Aerobic respiration** The process that leads to a complete oxidation of organic food in the presence of O_2 releasing CO_2 , H_2O and a large amount of energy.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + 686 \text{ kcal}$$
- Anaerobic respiration** Organic food is broken down incompletely to release energy in the absence of oxygen and the products are CO_2 , C_2H_5OH and $C_3H_6O_3$ (lactic acid).

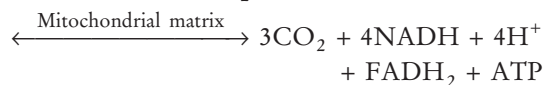
Glycolysis

The breakdown of a glucose molecule into two molecules of pyruvic acid is called **glycolysis**. Net yield of glycolysis is 6 ATP molecules.

TCA or Krebs's Cycle

TCA cycle starts with the condensation of acetyl group with Oxaloacetic Acid (OAA) and water to yield citric acid.

Pyruvic acid + $4NAD^+$ + FAD^+ + $2H_2O$ + ADP + Pi



$$\text{Respiration Quotient (RQ)} = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ absorbed}}$$

PLANT GROWTH

All living organisms show various changes in their weight, shape, size and volume during their entire life cycle (birth to death). The process of increase in the size, weight and shape is known as **growth**. The growth of plants is regulated by certain chemical substances, which are synthesised by them and these are called **growth hormones** or **growth regulators**. Plant growth regulators are also called as **phytohormones**.

Plant growth regulators are as follow:

Auxins

- Auxin was initially isolated from the urine of human, but later on its presence was also found in plants.
- They promote cell elongation, cell division and cell enlargement.
- IAA is natural auxin, while IBA, NAA and 2, 4-D are synthetic auxins. Auxins can inhibit lateral bud formation.

Gibberellins

- Gibberellins are weakly acidic growth hormone.
- It causes cell elongation and increases length and is produced in embryos, roots and young leaves near the shoot tip.
- It is helpful in flowering, enzyme synthesis and fruit growth.

Cytokinins

- Natural cytokinins are known to be synthesised in the regions, where rapid cell division takes place, e.g. root apex, young fruits, etc.
- They promote cytokinesis (cell division).
- Kinetin was first isolated from degraded sample of DNA.
- Zeatin was isolated from maize endosperm.
- It is also responsible for cell division, cell enlargement, prevention of senescence and enzyme synthesis.

Ethylene

- It is a gaseous hormone, which is produced from the ripening fruits and mainly acts as growth inhibitor.
- It fastens ripening of fruits and promotes ageing of plant organs.
- It promotes senescence and break dormancy.

Absciscic Acid (ABA)

- It is a growth inhibitor as it counteract other hormones.
- It is responsible for dormancy in buds and seeds, ageing in leaves, it inhibits mitosis, abscission of leaves, flowers and fruits.
- It causes partial closure of stomata and therefore, called **anti transpirant**.

REPRODUCTION

It is a biological process, in which organisms produce young ones (offspring) similar to themselves.

Reproduction is of two types:

- Asexual reproduction** When offsprings are produced by a single parent without the involvement of gametic fusion, e.g. unicellular (monerans and protists), many plants and simple animals.

Types of asexual reproduction are fission, budding, fragmentation, regeneration, vegetative propagation (potato), etc.

- Sexual reproduction** It is the process of development of a new individual and the formation and fusion of two gametes, e.g. all types of plants and animals.
- Flowers are the sit of sexual reproduction in angiosperms.
 - Androecium consisting of stamens represents the male reproductive parts. Gynoecium consisting of pistils represents the female reproductive parts.
 - Pollen grains (male reproductive organ) develop inside the microsporangia (anther).
 - The pistil has three parts—**stigma**, **style** and **ovary**. Ovules are present in the ovary.
 - Pollination is the mechanism to transfer pollen grains from the anther to the stigma.
 - A cell of the archesporium, the megaspore mother cell divides meiotically and one of the megaspore forms the embryo sac (the female gametophyte).
 - Angiosperms exhibit double fertilisation.
 - The mature embryo sac is 7-celled and 8-nucleated.
 - The developing embryo through different stages develops into seed.



IMPORTANT POINTS

- The study of structure and development of embryo is called Embryology. The great embryologist of India is **P Maheshwari**.
- **Tissue culture** It is the technique to exploit the property of totipotency of the plant cells.
- Rose, *Bryophyllum* and marigold can be propagated by cutting.

ECONOMIC BOTANY

- Natural rubber is para rubber, obtained from *Hevea brasiliensis*.
- Seeds of groundnut have 23-30% proteins.
- Starch is a polymer of glucose used as a thickening agent or emulsifier in pharmaceutical or detergents.
- Coir is a natural fibre obtained from fibrous mesocarp of coconut. It is used in products such as floormats, doormat's, etc.
- Sunflower is cultivated for oil and ornamental flowers.
- Red and black seeds of *Abrus* (Ratti) are used as jeweller's weight.
- *Eucalyptus* grows very fast and its stem is used in paper and pulp industry.
- Chief source of sugar in India is shoot of sugarcane (*Saccharum officinarum*).
- Coffee is obtained from seeds of *Coffea arabica* (family—Rubiaceae).

- Sunnhemp is the plant used for green manuring in India.
- Botanical name of tea is *Thea sinensis* (family–Theaceae).
- Most important cereals are rice, wheat, maize, etc.
- Banana, mango and *Citrus* are indigenous to India.
- Pungent smell in garlic is due to allicin compound.
- ‘Black gold of India’ is pepper.
- Cutting and peeling of onions bring tears to eyes because onion acids combine with sulphur to form amino acid sulphoxides.
- Quinine is a natural white crystalline alkaloid used in the treatment of malaria. It has antipyretic (fever reducing), antimalarial, analgesic and anti-inflammatory properties and found naturally in the bark of *Cinchona* tree.
- Bagasse is the fibrous matter remains after sugarcane stalks are crushed to extract their juice. It is used as a biofuel and as a substitute for wood in many subtropical and tropical countries for the production of pulp, paper, board, etc.
- Cereals are rich in carbohydrates and belong to family–Gramineae.

Common Medicinal Plants

Common name	Botanical name	Part used
Aconite	<i>Aconitum napellus</i>	Tuberous roots
Belladonna	<i>Atropa belladonna</i>	Leaves
Sarpagandha	<i>Rauwolfia serpentina</i>	Roots
Quinine	<i>Cinchona officinale</i>	Bark
Opium	<i>Papaver somniferum</i>	Fruits
Ashwagandha	<i>Withania somnifera</i>	Roots
Isabgol	<i>Plantago ovata</i>	Fruits and seeds
Ephedrine	<i>Ephedra</i>	Bark

ANIMAL PHYSIOLOGY

Animal physiology is the biological study of the functions of animals and their parts, including all physical and chemical processes. Physiology basically focuses principally at the level of organs and organ systems. Human physiology studies how our cells, muscles, organs work together and how they interact.

It involves growth and development, absorption and processing of nutrients, synthesis and distribution of nutritive substances and functioning of different tissues, organs and other anatomical structures.

NUTRITION

It is the sum of all those activities, which are concerned with ingestion, digestion and absorption of digested food into blood or lymph followed by egestion.

There are mainly two types of nutrition:

1. **Autotrophic or Holophytic nutrition**
Preparation of organic food from inorganic material by organism itself, e.g. all green plants, some unicellular organisms, e.g. like *Euglena*, *Chlamydomonas* and *Volvox* are autotrophs.
2. **Heterotrophic nutrition** Taking the readymade organic food material synthesised by autotrophs

is termed as heterotrophic nutrition, e.g. animals, parasites and saprophytes are heterotrophs.

Food or Diet is the nutritive substances taken by an organism for growth, work, repair and maintaining life processes. It is a mixture of various components, in which the quality and quantity of components may vary.

The major seven components of food are as follow:

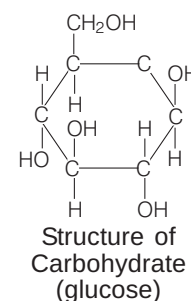
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| (i) Carbohydrates | (ii) Lipids |
| (iii) Proteins | (iv) Minerals |
| (v) Vitamins | (vi) Water |
| (vii) Roughage | |

Carbohydrates

Carbohydrates are polyhydroxy aldehyde or ketones, usually composed of C, H and O in the ratio of 1 : 2 : 1 with some exceptions. General formula of carbohydrates is $C_n(H_2O)_n$ (n = number of carbon atoms).

1g of carbohydrate = 17 kJ of energy.

Carbohydrates release energy more quickly as compared to fats.



Sources of Carbohydrates

Main sources of carbohydrates include potatoes, fruits, cereals (rice, wheat, maize), sugar, honey, bread, milk, etc.

Categories of Carbohydrates

On the basis of number of sugar units present, carbohydrates are of mainly following three types:

Monosaccharides

Definition	Examples	Functions
These are the simple carbohydrates, which are made up of only one sugar molecule.	Glucose, fructose, ribose, etc.	Glucose is the blood sugar, ribose, (5-carbon monosaccharide), forms the backbone of genetic molecule RNA.

Disaccharides

Definition	Examples	Functions
These are formed of two or more unit of monosaccharides inter-linked by glycosidic bonds.	Sucrose (sugarcane), maltose, lactose, etc.	Sucrose is the table sugar and used in preparation of desserts and can also act as a food preservative.

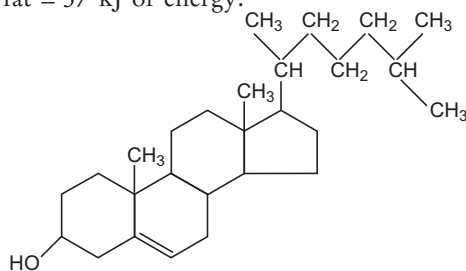
Polysaccharides

Definition	Examples	Functions
These are formed of large number (9<) of monosaccharides inter-linked by glycosidic bond.	Starch, glycogen, chitin, cellulose, hyaluronic acid, etc.	Polysaccharides serve for the storage of energy (e.g. starch and glycogen) and as a structural component (e.g. cellulose in plants and chitin in animals).

Lipids

Lipids are esters of long chain fatty acids and an alcohol called **glycerol**. They are insoluble in water, but soluble in non-polar solvents, e.g. chloroform, benzene or ether.

- Hydrolysis of fats occurs by a pancreatic enzyme called **amylase**.
- Animal fats are semi-solid, while vegetable fats are oils.
- 2 g of fat = 37 kJ of energy.



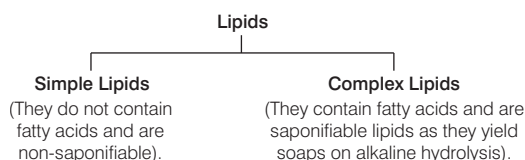
Structure of lipid

Sources of Lipids

Fats are supplied to our body by many food sources such as butter, ghee, cheese, milk, egg-yolk, meat, nuts, etc.

Categories of Lipids

On the basis of the presence or absence of fatty acids, lipids can be divided into two types :



Excessive intake of fat can lead to obesity and high cholesterol level gives rise to **hypercholesterolemia** resulting in cardiac damage. Fat deficiency results in drying of skin and deficiency of fat soluble vitamins. Fats are also required in cell membrane formation.

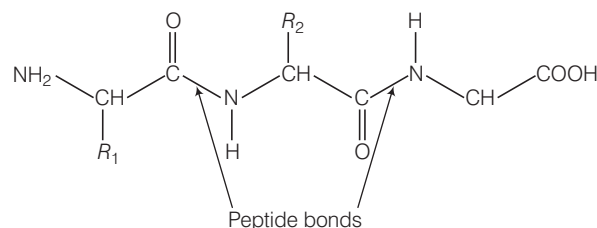
Sterols are also organic compounds. The most common sterol in animals is cholesterol.

The sterols found in plants are called as phytosterols, e.g. stigmasterol, sitosterol, etc.

Proteins

Proteins are the biochemical compounds consisting of one or more polypeptides folded in a typical way. Polypeptide is a single linear polymer chain of amino acids bonded together by peptide bonds.

- Proteins are composed of carbon, hydrogen, oxygen and nitrogen.
- Some proteins also contain sulphur, phosphorus and iron.
- There are 20 essential amino acids.
- The term proteins was given by J Berzelius (1938).



Structure showing peptide bonds between amino acids

Sources of Proteins

Main sources of proteins are cereals, pulses, fishes, eggs, milk, leafy vegetables, fruits, soyabean, peas, beans, cheese, curd, etc.

- Deficiency of proteins causes two diseases in children, i.e. **Marasmus** and **Kwashiorkor**.
- Silk is a protein fibre secreted by pupa of silk insect.

Some Essential Proteins

Body proteins	Functions
Enzymes	These are the biocatalysts, which help in biochemical reactions. All enzymes are proteinaceous in nature.
Hormones	These are non-virulent chemicals, which regulate body functions.
Transport proteins	Haemoglobin transports oxygen throughout the body.
Structural proteins	These help in the formation of cells and tissues.
Protective proteins	Antibodies help to fight against infection.
Contractile proteins	These proteins are responsible for contraction of muscles and movements, e.g. myosin, actin, etc.

Proteins are also found in the body of living organisms as enzymes. Thus, all enzymes are proteins only, but all proteins are not enzymes.

Minerals (Inorganic Salts)

The metals, non-metals and their salts are called **minerals**. They form about 4% of our body weight. Minerals are generally of two types:

- (a) **Macronutrients** (needed by body in large amount), e.g. calcium (Ca), phosphorus (P), potassium (K), sulphur (S), sodium (Na), chlorine (Cl) and magnesium (Mg).

- (b) **Micronutrients** (needed by body in small amounts), e.g. iodine (I), iron (Fe), zinc (Zn), manganese (Mn), copper (Cu), cobalt (Co), fluorine (F), molybdenum (Mo) and selenium (Se).

Sources of Minerals

These are obtained from the food and drink consumed by animals. These are also present in milk, eggs, meat, fruits, vegetables and table salts.

Sources and Functions of Different Minerals and their Associated Deficiency Diseases

Minerals	Daily requirement	Sources	Functions	Deficiency diseases
Sodium	Adequate amount in ordinary diet	Table salt, vegetables	Essential for nerve impulse conduction	Nervous depression, improper muscle contraction
Potassium	Adequate amount in ordinary diet	Vegetables	Required for muscle contraction and nerve impulse conduction	Nervous disorder, poor muscles leading to paralysis
Iron	18 mg	Leafy vegetables	Helps in oxygen transport as part of haemoglobin and myoglobin	Anaemia
Iodine	150 mg	Sea foods, water, onions	Essential for metabolic control as part of thyroid hormone thyroxine	Goitre
Calcium	1200 mg	Cereals, meat	Bone and teeth formation	Nervous disorder
Phosphorus	1200 mg	Meat, vegetables	Bone and teeth formation	Tetany and rickets
Magnesium	400 mg	Soyabean, green leafy vegetables	Activator of enzymes	Heart disease in infants
Zinc	15 mg	Beet, cheese	Cofactor of enzymes	Retardation of growth and loss of appetite
Manganese	5 mg	Peanuts	Cofactor of enzymes	Impairs glucose metabolism
Copper	Adequate amount in ordinary diet	Peanuts, beet	Cofactor of enzymes	Haematological and neurological disorder
Cobalt	Traces	Meat, milk	Part of vitamin-B ₁₂	Leads to deficiency of vitamin-B ₁₂
Fluorine	Traces	Water	Prevention of dental caries	Dental caries
Molybdenum	Traces	Water	Part of enzyme system	Neurological damage

Vitamins

Vitamins are the compounds essential for the metabolic activities of our body, but in very small quantities. These do not provide energy, but control energy yielding reactions of our body. The term 'vitamin' was coined by **Casimir Funk**.

On the basis of solubility, vitamins are of two types:

- (a) **Fat soluble vitamin**, i.e. Vitamin-A, D, E and K.
 (b) **Water soluble vitamin**, i.e. Vitamin-B complex and C.

Our body can synthesise vitamin-D and K.

Fat Soluble Vitamins

Fat in diet is essential otherwise these vitamins will not be absorbed.

Vitamin-A (Retinol)

- **Steenbock** (1919) discovered the vitamin-A and **Karrer** (1931) determined the structure of vitamin-A.
- It is also called as **antiinfective vitamin**.
- It is necessary for healthy eye sight (normal vision).
- It is destroyed by strong light.
- The main sources are yellow or green leafy vegetables (spinach), carrot, papaya, ripe mango, maize, milk, ghee, cod liver oil, etc.

- Deficiency causes night blindness (patient cannot see the object in dim light) and xerophthalmia or keratomalacia (dryness and wrinkleness of outer layer of eyeball).

Vitamin-D (Calciferol)

- It is called **poorman's vitamin** and is a sterol derivative.
- Its formation takes place under the skin in the presence of sunlight that's why also called as **sun-shine vitamin** or **antiricket vitamin**.
- It is needed for strong bones and teeth, helps in DNA synthesis, absorption of calcium and phosphorus.
- Some main sources are egg, milk, fish liver oil, etc.
- It affects the bones and causes rickets and osteomalacia in children and adults, respectively.

Vitamin-E (Tocopherol)

- It is also known as **beauty** or antisterility vitamin.
- It acts as an oxidant, helpful in making RBCs and is necessary for normal functioning of reproductive system in male and female both.
- The most important sources are green vegetable, wheat, vegetable oil and animal food.